

Java Basics & Pseudocode

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Exercise 1 – Personal Information

Solution Idea

The task is to print my name, age, and student identification. There are two simple steps to complete. First, declare variables for each. For the name, String is the ideal data type. For age, int suffices. Even though the student identification is a number, the String datatype is used. Since the student identification number is not intended for mathematical operations and may include leading zeros, it should be treated as a string rather than a numeric value.

Code

```

1  public class PersonalInfo 1usage 2Zoe
2  {
3      public static void runExcercise() 1usage 2Zoe
4      {
5          // Variable declaration section
6          String name = "Zoe Bunea";
7          int age = 20;
8          String studentId = "5209XXX";
9
10         // Using println method to print declared variables to the console
11         IO.println("My name is: " + name);
12         IO.println("My age is: " + age);
13         IO.println("My student ID is: " + studentId);
14     }
15 }
```

Figure 1: runExcercise method

```

public static void main(String[] args) 2Zoe
{
    PersonalInfo.runExcercise();
}
```

Figure 2: runExcercise being executed in main

Tests

```
My name is: Zoe Bunea  
My age is: 20  
My student ID is: 5209XXX
```

Figure 3: runExercise console output

Exercise 2 – Simple Calculator

Solution Idea

The task is to perform all simple arithmetic operations and display them. The result of each operation should be saved in their own variable. Since the task specifies that the input should be two integers, there's needs to be a cast to a float during the division process, to display the correct result.

Code

```
public class SimpleCalculator 2 usages 2 Zoe  
{  
    // Calculator takes two integer values as input  
    // and does simple calculations with them and prints it out to the console  
    public void simpleCalculations(int a, int b) 1 usage 2 Zoe  
    {  
        // Calculate sum, difference, etc and save them as variables  
        int sum = a + b;  
        int difference = a - b;  
        int product = a * b;  
        float quotient = (float) a / b; // cast to float to avoid integer division  
  
        // Print out the results  
        IO.println("Simple Calculator Results:");  
        IO.println(a + " + " + b + " = " + sum);  
        IO.println(a + " - " + b + " = " + difference);  
        IO.println(a + " * " + b + " = " + product);  
        IO.println(a + " / " + b + " = " + quotient);  
    }  
}
```

Figure 4: SimpleCalculator class

```
SimpleCalculator calculator = new SimpleCalculator();
```

Figure 5: SimpleCalculator Example Usage

Tests

```
Simple Calculator Results:  
4 + 3 = 7  
4 - 3 = 1  
4 * 3 = 12  
4 / 3 = 1.3333334
```

Figure 6: Simple Calculator Results

Exercise 3 – Turtle Maze

```
  0 1 2 3 4 5 6  
0 # E # # # # #  
1 # . . . # . #  
2 # . # . # . #  
3 # . # . # . #  
4 # # # . # . #  
5 # S . . . . #  
6 # # # # # # #
```

Figure 7: Turtle Maze Reference

Pseudocode

```
1    START  
2    TURN RIGHT  
3    FORWARD  
4    FORWARD  
5    TURN LEFT  
6    FORWARD  
7    FORWARD  
8    FORWARD  
9    FORWARD
```

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```
10  TURN LEFT
11  FORWARD
12  FORWARD
13  TURN RIGHT
14  FORWARD
```

Trace Table

Line	Statement	Row	Col	Direction
1	START	5	1	NORTH
2	TURN RIGHT	5	1	EAST
3	FORWARD	5	2	EAST
4	FORWARD	5	3	EAST
5	TURN LEFT	5	3	NORTH
6	FORWARD	4	3	NORTH
7	FORWARD	3	3	NORTH
8	FORWARD	2	3	NORTH
9	FORWARD	1	3	NORTH
10	TURN LEFT	1	3	WEST
11	FORWARD	1	2	WEST
12	FORWARD	1	1	WEST
13	TURN RIGHT	1	1	NORTH
14	FORWARD	0	1	NORTH

Exercise 4 – Number Guessing

Pseudocode

```
1    START
2    SET randomNumber = GENERATE RANDOMNUMBER(1, 100)
3    SET userGuess = 0
4    SET attempts = 1
5
6    PRINT "I'm thinking of a number between 1 and 100."
7
8    WHILE userGuess != randomNumber
9        PRINT "Enter your guess:"
10       READ userGuess
11
12       IF userGuess > randomNumber
13           PRINT "Too high! Try again."
14           INCREMENT attempts
15       ELSE IF userGuess < randomNumber
16           PRINT "Too low! Try again."
17           INCREMENT attempts
18       ELSE IF userGuess == randomNumber
19           PRINT "Correct! You guessed it in " + attempts +
           " attempts."
20    END
```

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Trace Table

For brevity reasons, most print statements have been skipped.

Line	Statement	userGuess	attempts	Condition	randomNumber
1	START	/	/	/	/
2	SET randomNumber = GENERATE RANDOMNUMBER(1, 100)	/	/	/	27
3	SET userGuess = 0	0	/	/	27
4	SET attempts = 1	0	1	/	27
8	WHILE userGuess != randomNumber	0	1	userGuess != randomNumber = true	27
10	READ userGuess	50	1	/	27
12	IF userGuess > randomNumber	50	1	userGuess > randomNumber = true	27
14	INCREMENT attempts	50	2	/	27
8	WHILE userGuess != randomNumber	50	2	userGuess != randomNumber = true	27
10	READ userGuess	25	2	/	27
12	IF userGuess > randomNumber	25	2	userGuess > randomNumber = false	27
15	ELSE IF userGuess < randomNumber	25	2	userGuess < randomNumber = true	27
17	INCREMENT attempts	25	3	/	27
8	WHILE userGuess != randomNumber	25	3	userGuess != randomNumber == true	27

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10	READ userGuess	27	3	/	27
12	IF userGuess > randomNumber	27	3	userGuess > randomNumber = true	27
15	ELSE IF userGuess < randomNumber	27	3	userGuess < randomNumber = false	27
18	ELSE IF userGuess == randomNumber	27	3	userGuess == randomNumber = true	27
19	PRINT "..."	27	3	/	27
20	END	27	3	/	27